

Answer **all** questions.

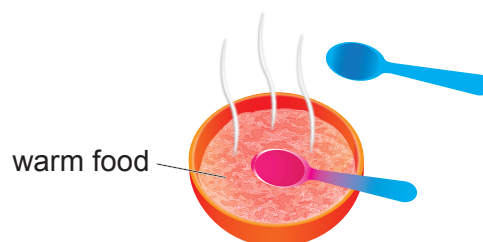
1. Smart materials are widely used in everyday life.

(a) The diagrams show some uses of smart materials which rely on their unusual properties.

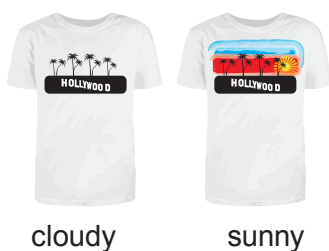
nappies



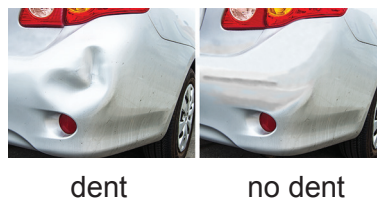
baby-feeding spoon



summer T-shirts



car bumper



The box contains the names of four different types of smart material.

thermochromic pigment

shape memory polymer

hydrogel

photochromic pigment



Use the information opposite to complete the table below.

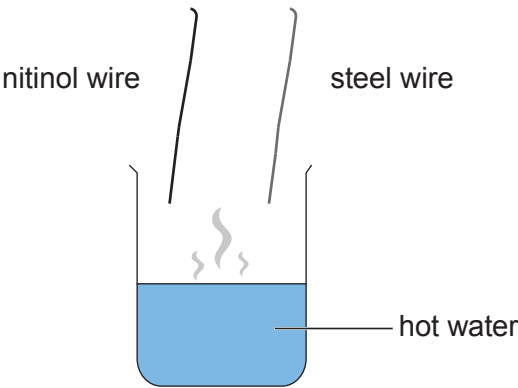
[4]

Unusual property	Use	Type of smart material
Able to absorb many times its own volume of water	nappies	hydrogel
Changes colour with changing temperature		
Changes colour with changing light intensity		

(b) Paperclips can be made from nitinol and from steel. Nitinol is a shape memory alloy but steel is not.



A student uncoiled both paperclips into straight wires and dropped them into hot water.



Put a **tick** (✓) in the box next to the statement that describes what happens.

[1]

- both wires reform into the shape of a paperclip

only the nitinol wire reforms into the shape of a paperclip

only the steel wire reforms into the shape of a paperclip

neither wire reforms into the shape of a paperclip
- ☐

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